MILITARY STANDARD

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

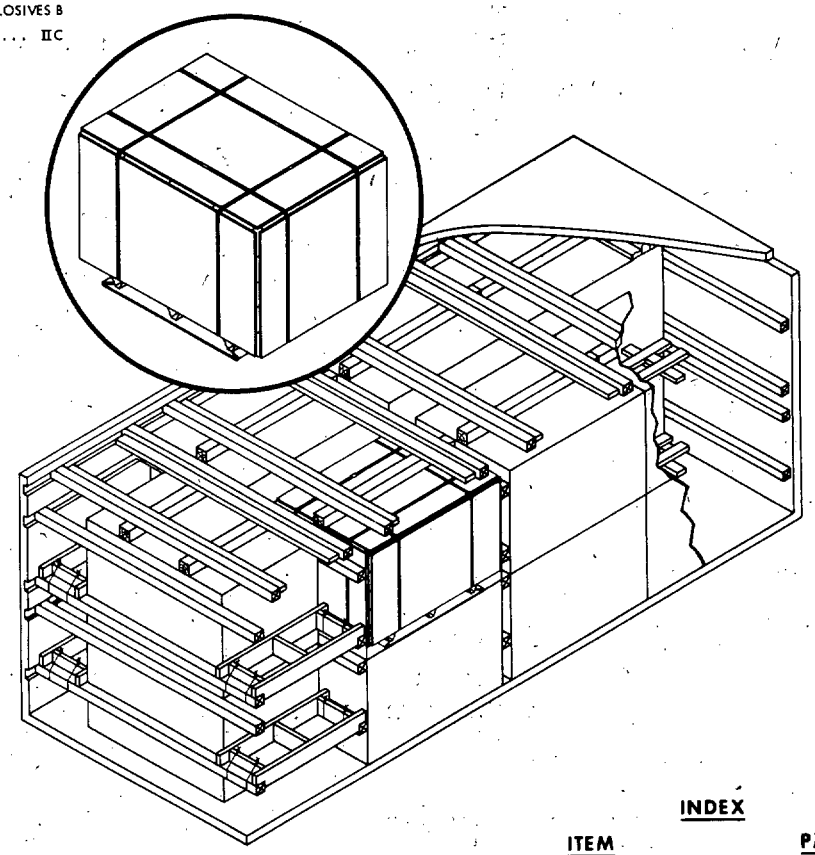
FLARE, AIRCRAFT PARACHUTE MK 24 & MODS
FLEET ISSUE UNIT LOAD

MIL-STD-1386-28
(NAVY)

1 OCTOBER 1973

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/154
DIMENSIONS ..... 57 5/8 L x 40 1/4 W x 38 H
CUBE ..... 51.0 CU. FT.
WEIGHT ..... SEE TABLE I
DOT HAZARD CLASSIFICATION .... EXPLOSIVES B
COAST GUARD CLASS IIC



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

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FSC 8140

AUTHORIZED FOR TRAILER-ON-FLAT-CAR (TOFC), CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS

M.S. Gray, NWHL — 9/14/73
SIGNATURE — TECHNICAL DIRECTION AGENT (TDA) — DATE

J.S. Kelly — 9/14/73
SIGNATURE — AIR SYSCOM BY DIRECTION — DATE

APPROVED BY US COAST GUARD

R.J. Donaly
DATE 31 Aug 1973

APPROVED BY BUREAU OF EXPLOSIVES

A.F. Grassmuck
DATE 9/4/73

ORIGINATOR L.R. Carnig — 9/14/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF AIRCRAFT PARACHUTE FLARES, MK 24 & MODS. THE FLARES SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/154.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.

CAUTION

WHEN POSITIONING CROSSMEMBERS, MAKE A VISUAL CHECK TO ENSURE THAT THE CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

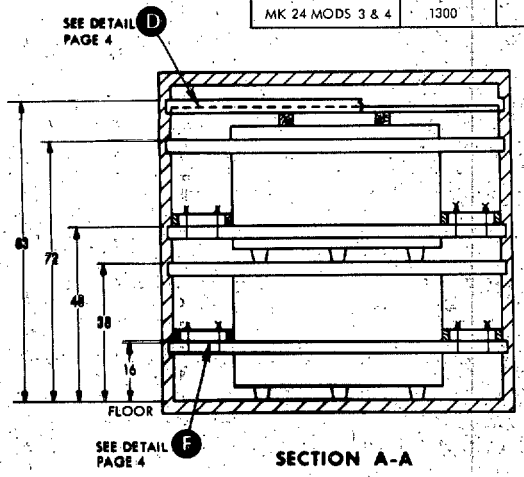
4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT.
5. WHEN LONGITUDINAL SLACK BETWEEN A CROSSMEMBER AND THE UNIT LOAD IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN CROSSMEMBER AND UNIT LOAD AS SHOWN IN DETAIL G.
6. THICKNESS OF HOLD DOWN ASSEMBLIES, DETAIL D AND E, MAY BE VARIED TO ACCOMODATE DIFFERENCES IN UNIT LOAD HEIGHTS BY VARYING THICKNESS OF SUPPORT PIECES 4 OR 6. STRUCTURAL INTEGRITY OF HOLD DOWN ASSEMBLY SHALL BE MAINTAINED. VOID BETWEEN HOLD DOWN CROSSMEMBER AND PIECE NO. 4 OR 6 OF HOLD DOWN ASSEMBLIES SHALL NOT EXCEED ONE HALF INCH.
7. SWAY BRACE, DETAIL C, IS ASSEMBLED IN PLACE. PIECE 1 IS POSITIONED INBOARD OF THE CORNER PALLET SKIDS. WITH PIECE 2 AGAINST PALLET SKIDS, NAIL TO PIECE 1 WITH THREE 8d NAILS PER JOINT. CUT PIECE 3 TO FIT BETWEEN PIECES 2 AND NAIL TO PIECE 1 WITH THREE 8d NAILS.
8. LENGTH OF STRUT, PIECE 8, SHALL BE SUFFICIENT TO CAUSE SWAY BRACE ASSEMBLY, DETAIL F, TO BEAR AGAINST UNIT LOAD AND CONTAINER SIDEWALL. TWIST TIE ASSEMBLY TO CROSSMEMBER WITH 16 GAUGE TIE WIRE FOUR PLACES.
9. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER - FED. SPEC. MM-L-751
   NAILS - FED. SPEC. FF-N-105, TYPE II STYLE 10, COMMON BRIGHT
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL. I).

TOFC/COFC, CONTAINERSHIP AND HIGHWAY

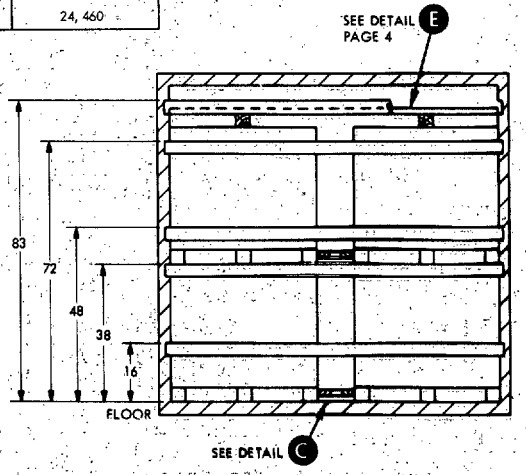
1. THE LOADED MILVAN CONTAINER SHALL BE ON A 20-FT MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES WHEN BEING MOVED IN TOFC SERVICE.
2. WHEN THE MILVAN CONTAINER IS SHIPPED TOFC OR COFC, THE RAIL CAR SHALL COMPLY WITH THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
3. THE LOADED CONTAINER MAY BE TRANSPORTED OVER PUBLIC HIGHWAYS ON A 20 FT MILVAN CHASSIS EQUIPPED WITH A DOUBLE BOGIE. SUITABLE CAPACITY COMMERCIAL EQUIPMENT MAY ALSO BE USED.

TABLE I

FLARE MK & MOD	UNIT LOAD WT. (LBS.)	CONTAINER LOAD WT. (LBS.)	CONTAINER GROSS WT. (LBS.)
MK 24 MOD 2	1180	16,520	22,870
MK 24 MODS 3 & 4	1300	18,200	24,460



SECTION A-A

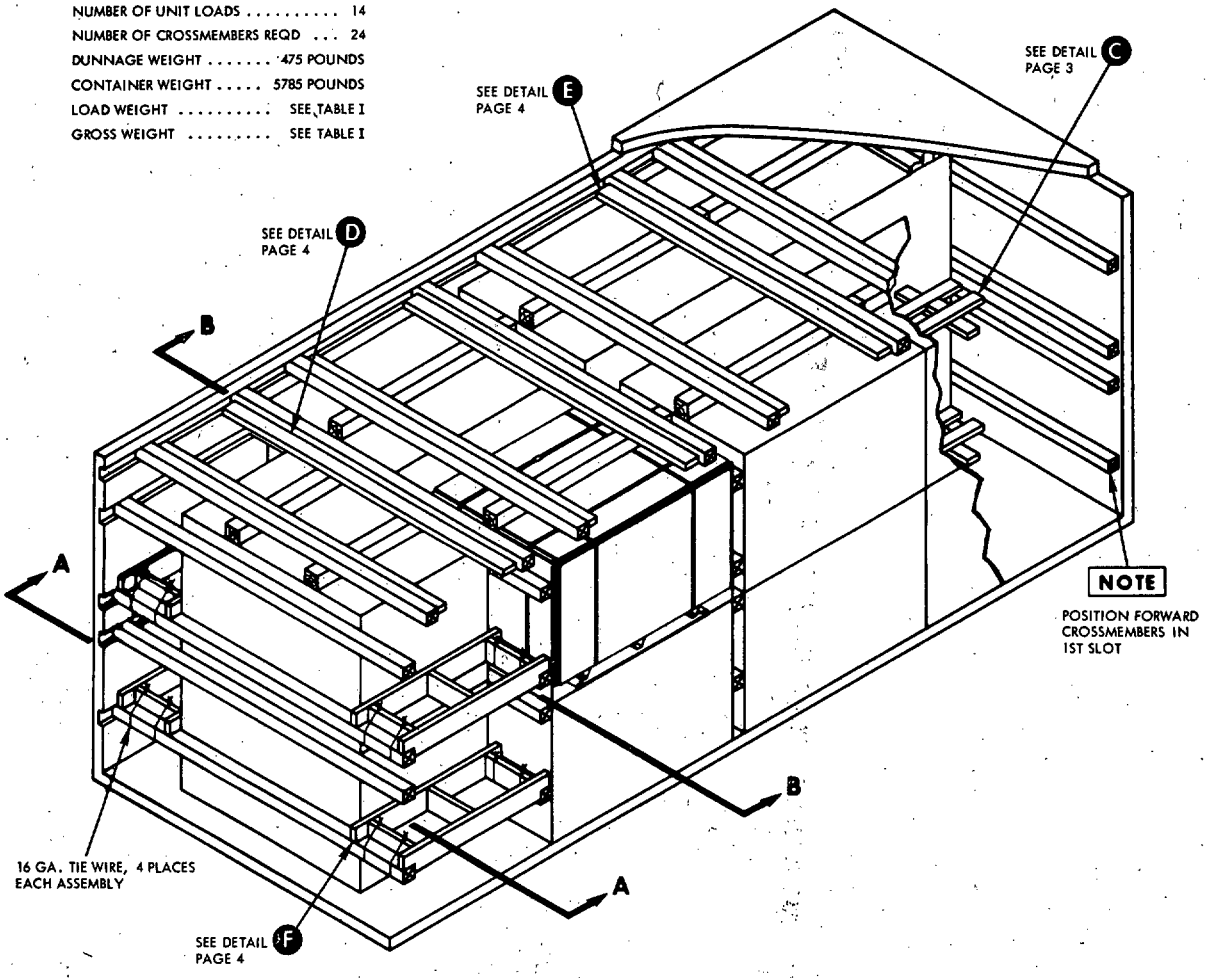


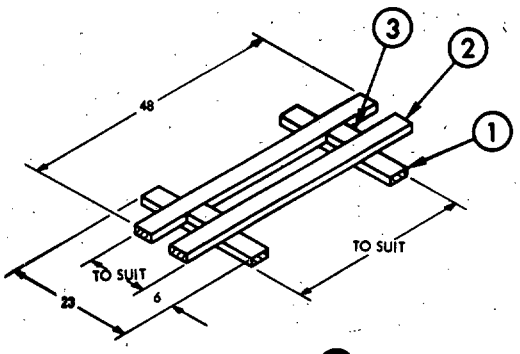
SECTION B-B

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CONTAINER LOAD DATA

NUMBER OF UNIT LOADS 14
NUMBER OF CROSSMEMBERS REQD . . . 24
DUNNAGE WEIGHT . . . . . 475 POUNDS
CONTAINER WEIGHT . . . . . 5785 POUNDS
LOAD WEIGHT . . . . . SEE TABLE I
GROSS WEIGHT . . . . . SEE TABLE I



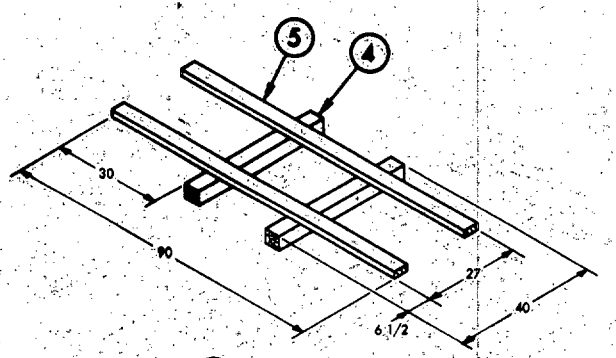


DETAIL C
SWAY BRACE ASSEMBLY
6 REQUIRED

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
9	STRINGER	2 x 6 x 46	8	8	3 PER JOINT	10d
8	STRUT	2 x 6 x TO SUIT	12	SEE 9	-	-
7	STRINGER	2 x 4 x 90	6	6	3 PER JOINT	16d
6	SUPPORT	4 x 4 x 56	6	SEE 7	-	-
5	STRINGER	2 x 4 x 90	2	4	3 PER JOINT	16d
4	SUPPORT	4 x 4 x 40	2	SEE 5	-	-
3	SPACER	2 x 4 x TO SUIT	12	1	3	8d
2	STRINGER	2 x 4 x 48	12	1	3 PER JOINT	8d
1	SUPPORT	2 x 4 x 23	12	SEE 2 & 3	-	-

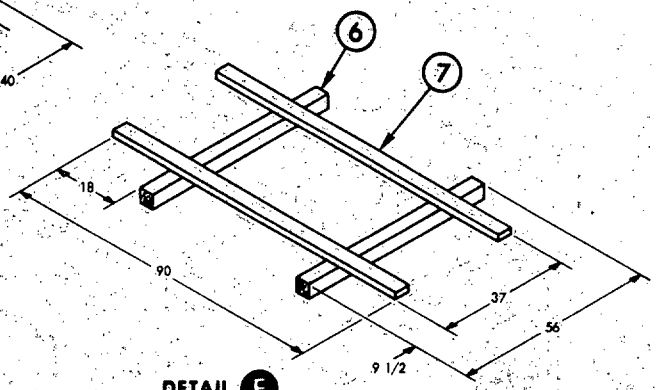
LIST OF MATERIALS AND NAILING DATA

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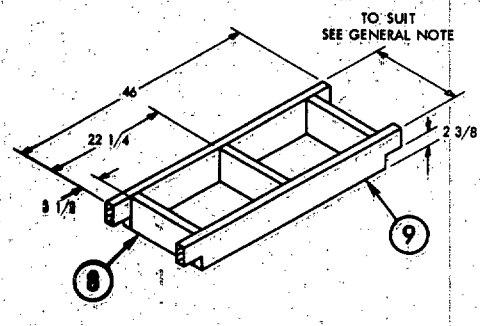
DETAIL D

HOLD DOWN ASSEMBLY
1 REQUIRED
SEE GENERAL NOTE 6



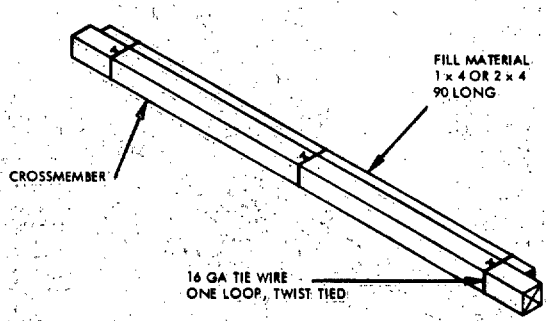
DETAIL E

HOLD DOWN ASSEMBLY
3 REQUIRED
SEE GENERAL NOTE 6



DETAIL F

SWAY BRACE ASSEMBLY
4 REQUIRED
SEE GENERAL NOTE 8



DETAIL G

FILL MATERIAL
SEE GENERAL NOTE 5

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